

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1706

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Rekha S K (192511023) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Rekha S K

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

- i. D1 cannot work Night shift ii. D2 must work before D3 (Morning < Afternoon < Night) iii. Only one doctor per shift iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1 iii. Cannot pass through obstacles iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

ANSWERS

1. HOSPITAL DOCTOR SHIFT ASSIGNMENT USING BACKTRACKING SEARCH

Variables

- D1
- D2
- D3

Domain

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

- $D1 \neq \text{Night}$
- D2 before D3
- One doctor per shift
- $D3 \neq \text{Morning}$
- Every doctor gets exactly one shift

Backtracking Search

Step 1

Assign **D1 = Morning**

Constraints:

- $D1 \neq \text{Night}$
- No conflicts

Current Assignment:

D1 = Morning

Step 2

Assign **D2 = Afternoon**

Constraints:

- Shift not already used
- D2 before D3 (to be checked later)

Current Assignment:

D1 = Morning

D2 = Afternoon

Step 3

Assign **D3 = Night**

Constraints:

- $D3 \neq \text{Morning}$
- Only one doctor per shift

- D2 before D3
 - Afternoon comes before Night

Current Assignment:

D1 = Morning

D2 = Afternoon

D3 = Night

Final Valid Schedule

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

2. CONSTRAINT-BASED PROBLEM SOLVING – ROBOT PATH PLANNING USING BFS

Problem Statement

A robot moves in a 5×5 grid from **Start (S)** to **Goal (G)**.

Constraints

1. Movement allowed:
 - Up
 - Down
 - Left
 - Right
2. Cost of each move = 1
3. Robot cannot move through obstacles.
4. Use **Breadth-First Search (BFS)**.

Algorithm:

Breadth-First Search (BFS) explores nodes level by level using a FIFO Queue.

S	.	.	X	.
.	X	.	X	.
.	X	.	.	.
.	.	X	X	.
.	.	.	.	G

Where

- S = Start
- G = Goal
- X = Obstacle
- . = Free Cell

BFS Node Expansion

Step	Queue	Expanded Node
1	S	S
2	(0,1), (1,0)	(0,1)
3	(1,0), (0,2)	(1,0)
4	(0,2), (2,0)	(0,2)
5	(2,0), (1,2)	(2,0)
6	(1,2), (3,0)	(1,2)
7	(3,0), (2,2)	(3,0)
...	...	...
Final	Goal Found	G

Optimal Path

S

↓

(1,0)

↓

(2,0)

↓

(3,0)

↓

(4,0)

→

(4,1)

→

(4,2)

→

(4,3)

→

G

Path Cost

Each move costs 1.

Total Moves = 8

Total Cost = 8

3. AUTONOMOUS RESCUE ROBOT

a) Analyze the given constraints and explain how they affect the robot's decision-making process.

Constraints and Their Effects

Constraint	Effect on Decision Making
Robot can move Up, Down, Left, Right	The robot has four possible actions from each position.
Obstacles cannot be traversed	The robot must avoid blocked cells and find an alternate route.
Limited sensing capability	The robot can only observe adjacent cells, so it must make decisions with incomplete information.
Cost of each movement = 1	Every movement increases the total path cost by 1.
Risky zones add an extra cost of 2	The robot should avoid risky areas whenever possible to minimize total cost.
Dynamic knowledge update	As new information is discovered, the robot updates its map and replans its path if necessary.

b) Solution Approach

Search Strategy Used

Online Search Agent with Uniform Cost Search (UCS)

Reason

- The environment is **partially observable**, so the robot does not know the complete map initially.
- The robot continuously gathers information while moving.
- Uniform Cost Search selects the path with the **minimum cumulative cost**, considering both movement cost and risky zones.

Steps

1. Start from the initial position **S**.
2. Observe all adjacent cells.
3. Ignore blocked cells.
4. Calculate the cumulative path cost.
5. Add an additional cost of **2** for risky cells.
6. Expand the node with the lowest total cost.
7. Update the environment whenever new cells are discovered.
8. Repeat until the robot reaches **G**.

c) AI Concepts Applied

1. Intelligent Agent

The rescue robot acts as an **Intelligent Agent** because it:

- Perceives the environment using sensors.

- Makes decisions based on available information.
- Plans a safe route.
- Learns new information while moving.
- Acts autonomously to rescue survivors.

2. Rationality

The robot is **rational** because it always chooses the action that:

- Minimizes total travel cost.
- Avoids obstacles.
- Avoids risky zones whenever possible.
- Safely reaches the survivor.

3. Environment Type

The rescue robot operates in the following environment:

Property	Type
Observability	Partially Observable
Determinism	Deterministic
Nature	Dynamic
State	Sequential
Representation	Grid Environment
Agents	Single-Agent

d) Justification

The **Online Search Agent with Uniform Cost Search (UCS)** is the most suitable approach because:

- It works well in **partially observable** environments.
- It updates knowledge dynamically as new information becomes available.
- It always selects the **least-cost path**.
- It considers both movement cost and additional risky zone cost.
- It ensures safe and efficient navigation to the survivor.

